# STAT 540 Midterm

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**Problem 1:(30 points)** Mark T(true) or F(false).

- (T, F) A binomial tree model is complete: TRUE
- (T, F) A conditional expectation of a random variable $X_t$ with respect to a $\sigma$-algebra G is a deterministic function : FALSE
- (T, F) The martingale probability ($q_u$,$q_d$) in a binomial tree model does not depend on the original probability ($p_u$,$p_d$) : TRUE
- (T, F) The Markov property tells that the future is independent of the past conditional on the current state: TRUE
- (T, F) A Brownian motion moves infinite distance in a finite time: TRUE

- (T, F) A Brownian motion has stationary and independent increments: TRUE

**Problem 2:(20 points)** A *binary* call option pays \$1 if the terminal stock price $S_T$ is at or above $K$, and it pays nothing otherwise. Similarly, a binary put option pays \$1 if $S_T \leq K$, and pays nothing if $S_T > K$. Let $\Phi_1(S_T)$ be the payoff function of a European call option with strike price $K$, and let $\Phi_2(S_T)$ be the payoff function of a binary call with strike price $K$.
Find a relation between $\Phi_2(S_T)$ and $\Phi_1(S_T)$.

Given:

$$\phi_1(ST) = (ST - K)^+ \ (European\ Call\ payoff)$$
$$\phi_2(ST) = 1\ if\ ST \geq K, 0\ if\ ST < K\ (binary\ call\ payoff)$$

Find the relationship:

$$\frac{\partial \phi_1(S_t)}{\partial k} = \frac{\partial (S_t - K)^+}{\partial k}$$

$$For\ S_t > K: (S_t - K)^+ = S_t - K,\ \ \frac{\partial \phi_1(S_t)}{\partial k} = -1$$
$$For\ S_t < K: (S_t - K)^+ = 0,\ \ \frac{\partial \phi_1(S_t)}{\partial k} = 0$$
$$For\ S_t = K: there\ is\ a\ jump\ discontinuity$$

$$Therefore: \phi_2(S_t) = -\frac{\partial \phi_1(S_t)}{\partial k}$$

**Interpretation:** A binary call can be approximated by a call spread: buying a call at strike K and selling a call at strike K+ε, then taking ε → 0 and scaling appropriately.

**Problem 3:(20 points)** Explain why a one-period trinomial model is not complete.

A market is complete if every contingent claim can be replicated by trading in the available assets. In a one-period trinomial model:

- States at maturity: 3 possible outcomes (up, middle, down)
- Available assets: 2 assets (stock and bond)

Why it's incomplete:

- Degrees of freedom: To replicate a derivative with 3 possible payoffs, we need to solve a system with 3 equations (one for each state).
- Number of variables: We have only 2 variables (Δ shares of stock, B bonds in the portfolio).
- This is a 3×2 system (3 equations, 2 unknowns).

Conclusion: In general, this system is overdetermined and has no solution. Therefore, not all derivatives can be replicated.
A market is complete if and only if the number of linearly independent assets equals the number of possible states.

**Problem 4:(40 points)** Consider a two-period binomial tree model. The initial stock price is \$100, and at each node of the tree, the stock $S$ can either jump up by a factor $u = 1.1$ or down by a factor $d = 0.9$. Assume that interest rate $R = 0$. Consider an option whose payoff at time $T = 2$ is given by

$$2(80 - S_T)^+ + (S_T - 90)^+.$$

(1) Find the price of the option at time $t = 0$.

First lets build the stock price tree:

| T | Leaf (Tree) | Value |
|---|---|---|
| 0 | S0 | 100 |
| 1 | S1(u) | 110 |
| 1 | S1(d) | 90 |
| 2 | S2(uu) | 121 |
| 2 | S2(ud) | 99 |
| 2 | S2(dd) | 81 |

Then we calculate the risk-neutral probability:

$$q = \frac{1 + R - d}{u - d} = \frac{1 + 0 - 0.9}{1.1 - 0.9} = \frac{0.1}{0.2} = 0.5$$

Then we calculate the terminal payoff at t=2:

$$V_2(uu) = 2(80 - 121)^+ + (121 - 90)^+ = 0 + 31 = 31$$
$$V_2(ud) = 2(80 - 99)^+ + (99 - 90)^+ = 0 + 9 = 9$$
$$V_2(dd) = 2(80 - 81)^+ + (81 - 90)^+ = 0 + 0 = 0$$

Then we calculate the terminal payoff at t=1:

$$V_1(u) = \frac{1}{(1 + R)} [qV_2(uu) + (1 - q)V_2(ud)] = [0.5 * 31 + 0.5 * 9] = 20$$

$$V_1(u) = \frac{1}{(1+R)}\,[qV_2(ud) + (1-q)V_2(dd)] = \,[0.5 * 9 + 0.5 * 0] = 4.5$$

Lastly we calculate the terminal payoff at t=0:

$$V_0 = \frac{1}{(1+R)}\,[qV_1(u) + (1-q)V_1(d)] = \,[0.5 * 20 + 0.5 * 4.5] = 12.25$$

2. Find values of the hedging portfolio at two nodes at time $t$ = 1.

At node (u) where S1 = 110:

$$Hedge\ portfolio\ (\Delta, B)\ must\ satisfy:$$

$$\Delta * 121 + B = 31\ (if\ up)$$

$$\Delta * 99 + B = 9\ (if\ down)$$

Solving:

$$\Delta(121 - 99) = 31 - 9$$

$$\Delta = \frac{22}{22} = 1$$

$$B = 9 - 99 = -90$$

Portfolio value equals 20.

$$At\ node\ (u): \Delta_1(u) = 1, B_1(u) = -90, V_1(u) = 20$$

At node (d) where S1 = 90:

$$Hedge\ portfolio\ (\Delta, B)\ must\ satisfy:$$

$$\Delta * 99 + B = 9\ (if\ up)$$

$$\Delta * 81 + B = 0\ (if\ down)$$

Solving:

$$\Delta(99 - 81) = 9 - 0$$

$$\Delta = \frac{9}{18} = 0.5$$

$$B = 0 - 81 * 0.5 = -40.5$$

Portfolio value equals 4.5.

$$At\ node\ (u)\colon \Delta_1(u) = 0.5, B_1(u) = -40.5, V_1(u) = 4.5$$

**Problem 5:(20 points)** Two fair coins are tossed.

(1) Write the $\sigma$-algebra generated by $\{(H,T)\}$.

$$The\ \sigma - algebra\ generated\ by\ \{(H,T)\} = \{\emptyset, \{(H,T)\}, \{(H,H),(T,H),(T,T)\}, \Omega\}$$

It consist of the empty set, the event itself, its complement, and the whole space.

(2) Let $X$ be the number of head. Find $E(X|\{(T,H),(H,H),(T,T)\})$.

$$X(HH) = 2$$
$$X(HT) = 1$$
$$X(TH) = 1$$
$$X(TT) = 0$$

$$This\ is\ a\ G-Measurable\ random\ variable\ equal\ to\ E[X|A]$$

On A (coin is fair, so each have a probability of 0.25):

$$E[X \cap A] = 2 * \frac{1}{4} + 1 * \frac{1}{4} + 0 * \frac{1}{4} = \frac{3}{4}$$

$$P(A) = \frac{3}{4}$$

$$E[X|A] = \frac{E[X \cap A]}{P(A)} = 1$$

On complement of A (coin is fair, so each have a probability of 0.25):

$$E[X|A^C] = X(H,T) = 1$$

Therefore E[X|G] = 1

Problem 6:(20 points) Let Xj be i.i.d and P(Xj = 1) = P(Xj = −1) = 1/2. Also, let $M_k = \sum_{j=1}^{k} X_j$.

(1) Show that Mk is a martingale.

Check for:

$$E[|M_{k+1}|] < \infty \; for \; all \; k$$

$$E[M_{k+1}|\mathrm{F}_k] = M_k$$

Integrability: Since |Xj| = 1, we have |Mk| <= k, so E[|Mk|] <= k, so it complains the integrability condition.

Martingale property:

$$E[M_{k+1}|\mathrm{F}_k] = E[M_k + X_{k+1} \mid \mathrm{F}_k] = E[M_k|\ \mathrm{F}_k] + E[X_{k+1}|\ \mathrm{F}_k]$$

Since Mk is Fk measurable then:

$$E[M_k|\ \mathrm{F}_k] = M_k$$

Since Xk+1 is independent of Fk:

$$E[X_{k+1}|\ \mathrm{F}_k] = E[X_{k+1}] = 1 * \frac{1}{2} + (-1)\frac{1}{2} = 0$$

So:

$$E[M_{k+1}|\mathrm{F}_k] = M_k + 0 = M_k$$

(2) Calculate $Var(M_k)$ and $[M,M]_k$, the quadratic variation. Discuss the difference between those two.

Variance:
Since Xj are independent:

$$Var(M_k) = Var\left(\sum_{j=1}^{k} X_j\right) = \sum_{j=1}^{k} Var(X_j)$$

$$Var(X_j) = E[X_j{}^2] - \left(E[X_j]\right)^2 = 1 - 0 = 1$$

$$Var(M_k) = k$$

Quadratic Variance:

$$[M, M]_k = \sum_{j=1}^{k} (M_j - M_{j-1})^2 = \sum_{j=1}^{k} (X_j)^2 = \sum_{j=1}^{k} 1 = k$$

Since Xj^2 = 1 always (whether Xj equals 1 or -1)

Discussion of difference:

$Var(M_k) = k$: This is the variance, which measures the expected squared deviation from the mean. It's a deterministic quantity: $E[(M_k - E[M_k])^2] = E[M_k^2] = k$.
$[M,M]_k = k$: This is the quadratic variation, the sum of squared increments. In this case, it equals k deterministically because each $|X_j| = 1$.

Key difference:

For general martingales, $[M,M]_k$ is path-dependent and can be random
$\text{Var}(M_k)$ is always deterministic (it's $E[M_k^2]$ when $E[M_k]=0$)
In this special case, they happen to coincide because $[M,M]_k = k$ is deterministic
For continuous martingales like Brownian motion: $\text{Var}(W_t) = t$ (deterministic) while $[W,W]_t = t$ is the realized quadratic variation

The quadratic variation tracks the realized volatility along a path, while variance tracks the expected squared deviation.